

Enterprise Microservices & Docker Troubleshooting Guide

1. Container Crash Loop Diagnosis (CrashLoopBackOff)

When Kubernetes reports CrashLoopBackOff or Docker containers exit immediately upon startup, execute the following commands:

```
# Check container logs
docker logs --tail 100 <container_id>

# Inspect exit status code
docker inspect --format='{{.State.ExitCode}}' <container_id>
```

2. Exit Code Reference

- Exit Code 137: Container killed due to Out of Memory (OOMKilled). Increase memory limits in deployment YAML.
- Exit Code 139: Segmentation fault in application process.
- Exit Code 1: General application exception or uncaught error.

3. Database Connection Pool Exhaustion

Symptom: HTTP 500 errors with 'Connection timeout waiting for idle object'.

Resolution: Increase max_connections in PostgreSQL and set pool_size=20 in HikariCP or SQLAlchemy config.